Learning Exploration with Agent-Aligned Map Memory in MiniGrid

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Abstract. PPO combines local perception with observation-derived map memory to explore partially observed maps, discover points of interest (POIs), and survive. An initial map underperformed mapless PPO, motivating agent-centered, heading-aligned readout with unchanged capacity and rewards. Across 4,800 episodes, aligned coverage is 89.2%, versus 68.5% original, 78.6% mapless, and 93.4% frontier. Excluding 30 maps previously used in diagnosis preserves the improvement. We explain the implementation choices and report their exploration, survival, and looping tradeoffs.

1 Problem formulation and environment design

The task is a finite-horizon partially observable Markov decision process. State s_t contains terrain, POIs, agent pose, current/previous monster position, time, and discovery/visit bookkeeping. The policy sees local o_t and deterministic memory $m_t = f(m_{t-1}, o_t, a_{t-1})$, rather than full state. It maximizes discounted return:

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], \qquad a_t \sim \pi_\theta(\cdot \mid o_t, m_t).$$

The reward is a proxy: evaluation separately measures coverage, discovery, and survival so a high return alone cannot establish successful exploration.

Geometry and difficulty. We extend Farama Foundation's MiniGrid [1] to reuse its tested grid, movement, rendering, visibility, and Gymnasium interface. A connected depth-first maze supplies dead ends requiring backtracking; carved rooms and extra openings add open areas, route alternatives, and opportunities to loop. This makes history useful without assuming the learned policy will exploit it. Connectivity and reachable POIs are checked to avoid measuring impossible exploration. POIs remain discoverable floor tiles, so navigation and search share the same traversable space.

Preset	Grid	POIs	Action budget T	Monster period
Easy	15×15	2	512	3 actions
Medium	21×21	3	1,024	2 actions
Hard	27×27	5	1,536	2 actions

The presets increase spatial demand and task load; Easy's slower patrol eases timing pressure. Larger budgets permit more travel and turning. Exact sizes, counts, periods, and budgets were fixed in the initial plan, not calibrated for equal difficulty or optimized floor-count ratios.

Threat and episode rules. Six-step spawn separation reduces immediate-contact starts without guaranteeing survival. The monster samples legal neighbors, excluding its previous cell when alternatives exist: a stochastic patrol creates timing decisions without a chase rule. Separate map/patrol RNG streams prevent behavior from consuming generation randomness. Contact is checked after agent movement and again after scheduled patrol movement, catching both collision paths. Contact terminates; the time limit truncates. POI completion leaves exploration running to reward remaining coverage. Seeds plus actions reproduce the episode.

2 Observation and action choices

MiniGrid's native wall-occluded 7×7 view preserves partial observability while providing nearby collision cues. Five one-hot channels (unseen, empty, wall, POI, monster) express categories without treating numeric object IDs as ordered magnitudes. Four actions—left, right, forward, wait—keep control focused on navigation; wait permits letting a visible patrol pass. Four heading indicators plus movement success, wall/monster contact, and remaining time form eight scalars. Movement success supports relative odometry; contact reports observable outcomes; remaining time distinguishes otherwise identical finite-horizon situations.

The seven-channel 55×55 map records observed cells, walls, floor, discovered POIs, visits, visible monsters, and the agent. A fixed canvas accommodates maps up to 27×27. Occupancy counts are clipped at ten and divided by ten to bound input, with ten a heuristic threshold. Old monster positions are cleared to avoid stale occupancy. Memory resets each episode; agent inputs exclude hidden terrain/POIs/threats and evaluator denominators.

3 Reward design and its tradeoffs

Sparse POI rewards alone give little guidance before a discovery. We therefore credit newly seen floor (ΔS_t) to encourage information gathering and first occupancy (ΔV_t) to reward actual travel; first POI sightings (ΔP_t) retain the explicit task objective:

$$r_t = 0.02\Delta S_t + 0.01\Delta V_t + \Delta P_t - 0.001 - 0.02\mathbf{1}_{\text{wall}} - \mathbf{1}_{\text{monster}}.$$

A small step cost discourages wasting the finite budget; wall contact penalizes blocked movement; death penalizes threat contact. One-time credits prevent reward from repeatedly revisiting cells or resighting POIs. Reset initializes history without paying it, avoiding reset bonuses. Turns can earn information reward only for genuinely new observations. No reward clipping or normalization obscures these scales.

The weights are transparent heuristics, not tuned optima or safety guarantees. For a 236-cell Medium map, complete revelation and visitation offer about 4.7 and 2.4 reward respectively before subtracting reset credits; three POIs offer three. Death's immediate penalty equals one POI and also removes future reward opportunities. Thus the scale permits substantial exploration reward alongside death, motivating separate survival reporting. The legacy no-bonus ablation tests the contribution of seen/visited shaping in the original frame; it cannot establish the best weights or their interaction with alignment.

4 Why this memory, network, and RL algorithm?

Spatial history and alignment. Identical-looking junctions may require different actions after different branches have been explored. Reliable movement feedback allows an inspectable map instead of requiring the policy to learn localization from reward. Initial diagnostics found correct mapping but coordinate sensitivity: translating all map channels four cells changed 13.5% of greedy decisions; jointly rotating map and heading changed 20.5% across 864 sampled states. These distribution-shifting probes motivated a frame hypothesis, rather than proving its cause.

The policy now reads memory with the agent centered at $(27, 27)$ and forward upward. For output column u, row v, agent position p, and forward/right unit vectors f_h, r_h, it gathers

$$p_{\text{src}} = p + (27 - v)f_h + (u - 27)r_h.$$

An east-facing agent's output $(27, 26)$ reads one cell east and $(28, 27)$ one cell south. This aligns history with relative actions, reducing the coordinate conversion the network must learn. Integer indexing preserves symbolic values; zero padding avoids wrapping. Pose uses observable heading and the map marker. All seven channels transform together; storage, rewards, and zero-map inputs retain their meanings.

Component	Layers	Output
Local CNN	3×3, $5 \to 16 \to 32$, stride 1; flatten and linear	64
Map CNN	3×3, $7 \to 32 \to 64 \to 64$, stride 2; flatten and linear	128
Fusion	Local + map features + 8 scalars	200
Actor / critic	Separate $200 \to 128 \to 128$ MLPs; logits / value	4 / 1

Separate CNNs retain immediate tile detail while compressing historical context. Convolution shares filters across recurring geometry; scalar concatenation avoids treating time and contact as images. Shared encoders avoid duplicating perception, while separate heads estimate action probabilities and value. CNN layers/projections use ReLU and the SB3 heads use Tanh. Widths, striding, activations, and feature allocation are engineering defaults for this implementation, not architecture-search results. All variants have 649,925 parameters; 459,008 belong to the map encoder. Alignment adds none. This feed-forward network predicts actions, not explicit search plans; temporal history resides in the map.

PPO and resource fit. PPO [3] in Stable-Baselines3 [2] supports discrete actions, dictionary inputs, custom CNNs, and parallel rollouts. Its clipped probability-ratio objective, GAE value estimates, entropy bonus, target-KL stopping, and gradient clipping provide practical optimization controls. PPO reuses and then discards short on-policy rollouts. A later resource check strengthens its practical fit: our float32 input is 85,712 bytes, so one million observation/next-observation pairs in standard SB3 dictionary replay require about 171 GB before other fields; 4,096 rollout observations occupy about 335 MiB. This is a later feasibility observation, not the original selection rationale or proof against compressed DQN replay.

5 Training choices, controls, and baselines

Training mixes Easy and Medium equally per reset to avoid specializing to one scale; Hard is deliberately withheld to test extrapolation beyond training. Map seeds 0–9,999 provide varied layouts. Seeds 11, 22, and 33 provide repeated optimization runs rather than a single favorable result. Four conditions isolate intended factors: full-reward original map; full-reward zero map; original map without seen/visited bonuses; full-reward aligned map. Zero-map PPO still maintains discovery history for rewards, so it removes policy memory input rather than changing reward accounting.

Each run targets five million transitions, approximately 60 million over 12 runs. This is a matched compute budget, not a convergence claim. The three aligned runs each finish 5,001,216 transitions because PPO completes rollouts. Python 3.11 and an RTX 4090 provide the training runtime. Original-map seed 33 resumed after a CUDA failure, restarting sampling; a seed-11/22 sensitivity check addresses that reference-run concern.

Setting	Value and practical role
Collection	16 subprocess environments × 256 steps: 4,096 transitions/update; parallel sampling and bounded rollout storage
Updates	Batch 256, four epochs, learning rate 3×10^{-4}: minibatch reuse with limited update work
Return estimation	$\gamma = 0.99$, GAE $\lambda = 0.95$: discount distant rewards and trade estimator variance against bias
Update controls	Clip 0.2, entropy 0.01, target KL 0.03, value coefficient 0.5, gradient norm cap 0.5

These are the fixed settings from the initial plan; several differ from SB3 defaults, and no search establishes their optimality. Entropy encourages action diversity during training; clipping, KL monitoring, and gradient clipping constrain updates without guaranteeing stability. The scale $1/(1-\gamma) = 100$ steps highlights discounted emphasis on nearer rewards; it is not a hard planning horizon. A contribution to long-episode stalls remains an untested hypothesis.

Every 250,000 transitions, CPU validation uses Easy/Medium seeds 10,000–10,049. Selecting by coverage AUC rewards both early information gain and sustained final coverage, rather than shaped return alone; POI fraction and survival break ties. Keeping selection off test maps avoids choosing fortunate test checkpoints. Aligned selected steps are 3.25M / 4.50M / 3.75M for seeds 11 / 22 / 33. Mean selected/final validation AUC is 0.883/0.867, versus 0.715/0.515 for the original map, showing less late regression in these runs.

Why these references? Uniform seeded random actions establish a minimal exploration floor, while explicit one-time-reward tests address reward exploits. Deterministic frontier search is a strong classical coverage reference using the same observed map: BFS routes to known cells adjacent to unknown space, with fixed tie-breaking and visible-threat avoidance. It is not an upper bound. Keeping map inputs, budgets, and threat rules comparable prevents a hidden-information advantage; ablations deliberately remove one component. The mapless comparison also changes active optimization capacity because its map branch receives zeros.

6 Evaluation methodology and seed disclosure

The archived comparison uses seeds 30,000–30,099 on every preset. **Overlap disclosure:** an earlier review evaluated old original-map and mapless policies on 30,000–30,029 and informed the memory investigation. Earlier seeds 20,000–20,099 were also exposed. Consequently, the full 100-map comparison is not wholly independent architectural confirmation, even though aligned checkpoints were selected on validation only. We retain all recorded results and additionally exclude those 30 maps as a post-hoc sensitivity analysis.

Each PPO condition contributes $3 \times 100 \times 3 = 900$ episodes; random contributes 900 over three policy seeds and frontier 300, totaling 4,800. PPO is greedy, frontier deterministic. Deterministic evaluation fixes action choice for reproducibility, while its tendency to stall is reported separately. No aligned checkpoint was chosen using this comparison.

Revealed/visited coverage divide observed/occupied floor by evaluator-only traversable-cell counts. POI fraction measures discovery; survival means reaching the time limit. A loop is a position-heading repetition within 16 prior states with no new floor or POI; the 16-state window is a diagnostic convention. Coverage AUC is normalized trapezoidal area over the planned budget. Curves hold terminal coverage/discovery after death to avoid survivor-only averages, with survival reported separately. Presets receive equal weight, preventing different budgets from dominating. Episode SD and SD across three run means describe different variation; paired-map bootstrap intervals average runs within maps and resample maps. Their 95% intervals condition on these models and are not multiplicity-corrected.

7 Results, sensitivity, and generalization

Table 1: Revealed coverage (%), 100 comparison seeds per preset, including the disclosed 30-map overlap. ± is SD of three run means (policy seeds for random); frontier is deterministic.

Agent	Easy	Medium	Hard
Random	65.6 ± 0.6	41.1 ± 1.4	27.1 ± 0.6
Frontier	93.9	93.4	92.7
PPO, no bonuses (original frame)	70.2 ± 2.3	40.6 ± 2.0	22.3 ± 0.8
PPO, no map	92.6 ± 1.7	79.4 ± 2.0	63.9 ± 4.3
PPO, original map	87.6 ± 3.2	69.8 ± 5.3	48.2 ± 8.5
PPO, aligned map	96.8 ± 1.1	90.4 ± 2.0	80.4 ± 3.4

Aligned PPO gains 20.7 percentage points (pp) over the original map and 10.6 pp over mapless PPO on equal-preset coverage. Visited coverage and POI discovery improve too (Table 2), showing actual travel as well as visibility gain. Aligned Hard coverage has episode SD 21.0 pp versus run-mean SD 3.4 pp; three runs still give limited evidence about optimization variability.

Table 2: Equal-preset means on all 100 maps. Percentages except AUC (0–1). Coverage gathered before death remains credited; survival is a separate outcome.

Agent	Coverage	Visited	AUC	POIs	Survival	Loops
Random	44.6	11.2	0.396	46.4	28.7	68.1
Frontier	93.4	60.8	0.886	94.0	37.7	21.0
PPO, no bonuses	44.4	10.5	0.399	45.6	60.9	85.2
PPO, no map	78.6	48.1	0.697	80.6	76.7	32.4
PPO, original map	68.5	36.2	0.617	67.5	72.9	45.7
PPO, aligned map	89.2	63.3	0.814	89.5	70.9	41.4

Overlap and resume sensitivity. On seeds 30,030–30,099, coverage is 89.5% aligned, 78.1% mapless, and 68.5% original. Paired aligned-minus-mapless coverage gains and 95% map-bootstrap intervals (pp) are Easy +3.8 [1.7, 6.0]; Medium +12.5 [9.6, 15.4]; Hard +18.0 [14.0, 22.1]. This post-hoc exclusion preserves the result without converting the study into a preregistered confirmation. Separately excluding resumed original seed 33 retains aligned-minus-original gains of +8.2 / +18.7 / +25.4 pp using matched seeds 11 and 22 on all 100 maps.

Reward and safety. Original-frame no-bonus PPO reaches 44.4% coverage, near random's 44.6%, supporting shaping in that frame. Frontier leads overall coverage and AUC. On Hard, aligned PPO trails frontier by 12.3 pp coverage but survives more often (64.7% versus 37.0%). Against mapless PPO, aligned Hard survival instead falls 11.7 pp [95% interval: -19.0, -4.3]. The representation improves exploration, with a safety tradeoff.

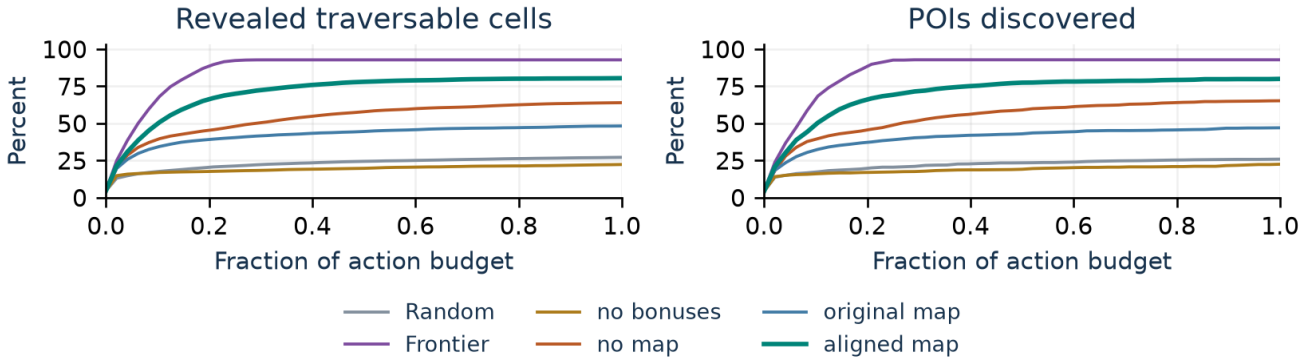


Figure 1: Hard coverage and POI discovery on the full comparison set. Terminal values persist after death. POI gains track greater exploration; these curves do not isolate a specialized POI-search strategy.

Extrapolation. Hard coverage falls 10.0 pp from Medium for aligned PPO, versus 21.5 pp for the original map, 15.5 pp for mapless PPO, and 0.7 pp for frontier. Hard changes size, POIs, budget, and agent-relative spatial extent together, so the drop cannot isolate one cause. Against seeds 0–49 from the training partition, aligned comparison-set coverage differs by -1.2 / -1.6 / -0.7 pp (Easy/Medium/Hard). Those seed numbers need not have been sampled during training, and no Hard maps were trained on. Comparable Easy/Medium comparison-set AUC is 0.865; its three-preset 0.814 also includes Hard and should not be directly compared with validation.

8 Failure cases and their implications

Survival without exploration. On Hard map 30029, aligned training seed 11 survives all 1,536 actions but reveals only 10.0% of terrain, discovers no POIs, and loops on 99.9% of actions. This post-hoc extreme shows why survival alone is inadequate and why greedy policies need incomplete-coverage diagnostics. The map supplies history but does not force the policy to act on it.

Early threat contact. On Hard map 30002, the same run dies after 32 actions with 8.2% coverage and one POI. Current-visibility-only threat memory prevents stale occupancy from masquerading as current truth, but loses threat tracking under occlusion. The example motivates testing predictive threat history; it does not identify the unique cause of contact.

Aligned Hard loops affect 49.7% of actions, versus 34.6% mapless and 46.5% original. Some repeats occur after useful exploration is complete, so the coverage context matters. Wall contacts average 1.56% of all aligned actions versus 0.47% mapless; this does not isolate errors per forward attempt. Higher exploration may increase threat exposure, but a causal attribution requires a separate test.

9 Computational feasibility and reproducibility

Isolated CPU timing uses one process/thread, two shared validation maps per preset, and 96 episodes. Action selection plus input preprocessing includes the first action but excludes physics, rendering, and loading, separating agent cost from simulation. Aligned PPO averages 1.204 ms (p95 1.427), versus 0.884 ms original, 0.891 ms mapless, and 0.523 ms frontier. Alignment adds gather cost without parameters; the map encoder remains 70.6% of capacity. These timings establish feasibility on the measured machine, not hardware-independent guarantees.

The MiniGridRL repository retains Farama history and Apache-2.0 attribution. Configurations, versioned/hashed checkpoints, raw measurements, training histories, and CPU/UI instructions support reproduction; aligned models are in `checkpoints/aligned/`. Automated checks cover determinism, information boundaries, rewards, geometry, and loading. The refreshed demo and Windows CPU build use validation-best aligned seed 33 (AUC 0.889863); historical artifacts remain archived separately.

10 Remaining limitations and next experiments

Rationale versus validation. Task structure motivates spatial history, relative alignment, and local/context separation; controlled evidence supports alignment and legacy shaping. Exact CNN widths, feature split, activations, sharing, and visit saturation lack comparative ablations. Reward weights, geometry/spawn/patrol constants, and budgets are heuristics, not calibrated optima. Five million transitions and three seeds provide matched cost and replication, not convergence or statistical power guarantees.

PPO fits the implementation and resource constraints, but no DQN or recurrent-policy comparison establishes algorithm superiority. Reward scale and γ could influence stalls and survival; no weight/discount sweep establishes either explanation. The mapless condition changes both information and effective learning capacity. Rotation probes also modify heading and are out-of-distribution, so the retrained frame comparison is stronger evidence than perturbation sensitivity alone.

What to test next. Train aligned PPO without seen/visited bonuses to measure reward–representation interaction. Then independently vary encoder size, visit saturation, reward/death balance, or discounting with fixed budgets and validation selection. Compare sampled actions and threat-history/recurrent policies on stalls and contact. Add training seeds and independently reserved maps. Hard's larger relative extent is a joint extrapolation challenge. Prior map exposure, post-hoc sensitivity, and extreme examples remain disclosed; further selection requires new confirmation maps.

Conclusion. Relative alignment made the same PPO network's map more useful for exploration, including after overlap exclusion. Frontier's stronger coverage and PPO's stalls and survival losses delimit the finding. Task-based motivation supports the design; exact constants still need controlled validation.

References

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